
Low-profile Wideband 1×2 Array Antenna for IEEE 802.11a/h/j/n/ac/ax WLAN Applications

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Abstract

In this paper, a novel low-profile wideband 1×2 array microstrip patch antenna has been proposed for wireless local area network (WLAN) applications. The antenna is comprised of a microstrip feed line and a directly coupled T-shaped junction, which is connected to circular patches through sub-feeders. The material of the substrate is Rogers RT5880. The ground plane is nearly half the size of the antenna. The antenna has an overall dimension of $40 \times 22.4 \times 0.79 \text{ mm}^3$. The antenna is developed step-by-step to evaluate gain and radiation pattern characteristics for optimum performance. The antenna is designed, simulated, and analyzed using CST Microwave Studio. The simulated outcomes are consistent. This antenna is suitable for IEEE 802.11a/h/j/n/ac/ax WLAN applications due to its operating frequency range of 4.72-6.32 GHz and

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center frequency of 5.672 GHz. Therefore, the simulated verification indicates the proposed antenna could be a suitable choice for WLAN applications.

Keywords— Low-profile antenna; microstrip patch; Wideband; Array; CST; WLAN.

1 Introduction

Institute of Electrical and Electronics Engineers (IEEE) 802.11 standards devices have created a market with a significant impact on the household, workplace, and public areas. Numerous devices, including computers, PCs, printers, cell phones, VoIP phones, media players, and Blu-ray players, utilize wireless LAN technology today [5]. IEEE 802 is a set of standards developed by the IEEE for local area networks (LAN), personal area networks (PAN), and metropolitan area networks (MAN). The IEEE 802 LAN/MAN standards committee (LMSC) is responsible for maintaining these specifications. In 1999, the IEEE 802.11a standard was published. Networks utilizing 802.11a operate at 5 GHz or 3.7 GHz radio frequency and 20 MHz bandwidth. The specification employs a modulation technique recognized as orthogonal frequency-division multiplexing (OFDM), which is ideally adapted for workplaces. With 802.11a, data transfer rates of up to 54 Mbps are feasible.

In 2009, the 802.11n standard was approved; it is implemented in multiple wireless antennas to transfer and receive information. OFDM is the modulation technique utilized by the IEEE 802.11n standard. Multiple input, multiple output (MIMO) describes the antenna technology employed by the IEEE 802.11n standard. The 802.11n networks use radio frequencies of 2.4 GHz or 5 GHz as well as a 20 MHz or 40 MHz channel width. The maximum possible network data transfer rate supported by 802.11n is 300 Mbps. The indoor and outdoor ranges for IEEE 802.11n are 75 m and 250 m, respectively [1]. The IEEE standard 802.11h, which adjusts MAC and PHY, defines procedures for dynamic frequency selection (DFS) and transmit power control (TPC) for wireless LAN devices. Using the same 5 GHz frequency band creates management issues, such as disturbances with satellite systems and radar. It was originally developed to comply with European regulations but is currently applied to a great number of other nations. Dynamic frequency selection (DFS) and transmit power control (TPC) are provided by this standard for the 802.11a PHY. It has subsequently been incorporated into the complete IEEE 802.11-2007 specification [15].

With the IEEE 802.11j amendment, wireless communication for 4.9 GHz as well as 5 GHz bands is extended in Japan. IEEE 802.11j enables communications for wireless local area network (WLAN)-compliant mobile and outdoor/indoor applications. This amendment outlines defined changes to the medium access control layer (MAC) and physical layer (PHY) of IEEE Stan-

standard 802.11 that permit a minimum speed of 10 Mbps and a maximum speed of 9.6 Gbps, as determined just at the MAC data service access point (SAP). The fifth iteration of WiFi standards, IEEE 802.11ac, was introduced in December 2013. This standard does have a bandwidth of 20, 40, 80, and 160 MHz and operates at 5 GHz. The ranges of stream rates for these bandwidth sectors are as follows: 65–866.7 Mbps for 160 MHz, 32.5–433.3 Mbps for 80 MHz, 15–200 Mbps for 40 MHz, and 7.2–96.3 Mbps for 20 MHz. This standard provides superior performance and coverage in comparison with IEEE 802.11a, b, g, and n. The 802.11ac standard employs a broader channel and an enhanced modulation scheme to support a greater number of clients. Multi-user MIMO is the technique utilized by the IEEE 802.11ac standard. This technique enables communication between a group of consumers or wireless platforms, each of which has one or several antennas. The indoor capacity is 35 m, and the highest outdoor range has not been recorded yet [1].

The year 2019 saw the introduction of IEEE 802.11ax standard. The 802.11ax networks operate at frequencies of 2.4 GHz and 5 GHz. The specification employs a modulation technique defined as orthogonal frequency-division multiplexing (OFDM), which is particularly well-suited for use in 802.11ax, where data transfer rates of up to 3.4–14 Gbps are feasible. The IEEE 802.11ax is a developmental protocol intended to enhance the performance of 802.11ac, which has minimal link contention-based access. 802.11ax offers increased network capacity, increased efficiency, enhanced performance, and decreased latency. The primary objective of this modern protocol is to implement new methodologies to serve too many users with a consistent and accurate flow of data that aims to boost throughput [17]. On the other hand, array antennas have been extensively used in modern antenna advancements to improve the efficiency of communications networks [7].

In this work, a low-profile ($40 \times 22.4 \times 0.79 \text{ mm}^3$) wideband 1×2 array antenna is designed and analyzed for IEEE 802.11 a/h/j/n/ac/ax WLAN applications. The compact antenna has a wide bandwidth of 1.60 GHz (4.72–6.32 GHz), a peak gain of 4.705 dB, and a maximum efficiency of 89.90%, which make it suitable for the intended applications. It also possesses a VSWR of 1.01 and a reflection coefficient of -45.5 dB at 5.672 GHz. The paper is further structured as follows: A literature review has been provided in Section 2; the design structure of the array antenna is depicted in Section 3; and the results and analysis are discussed in Section 4. At last, significant findings are made, and the paper is concluded in Section 5.

2 Literature Review

Due to the rapid development of short-range radio networks, including Bluetooth (BT) and wireless local area network (WLAN), in the cellphone com-

munications industry, the demand for antennas with compact dimensions and multiband operations is increasing. In WLAN IEEE 802.11, 2.4 GHz, 5.2 GHz, and 5.8 GHz are the optimal frequency bands for WLAN applications that a single antenna should encompass. For such applications, numerous WLAN antennas have been recently proposed and documented in the literature [18]. A 1×4 patch array antenna for wireless LAN applications is therefore introduced in [7]. Its structural dimension is $210 \times 150 \times 1.6 \text{ mm}^3$, and its substrate material is felt. The frequency coverage is 5.35–6.46 GHz, and the return loss is about -28 dB at about 5.8 GHz. A microstrip patch antenna is the preferred option for a number of applications due to its straightforward design and low-profile operation. It has a number of drawbacks, including a decrease in gain, a restricted bandwidth, and decreased efficiency because of surface and metallic wave degradation. In [8], the authors have discussed the development of dielectric resonator antenna arrays for WLAN, radar, direct broadcast satellite, and GSM 1800 applications. It has a volume of $50 \times 40 \times 1.6 \text{ mm}^3$. It possesses a maximum gain of 2.66 dB, and the return loss at center frequency is -20 dB. Two low-profile monopole multiband antennas are presented in [3]. The antennas consist of a T-shaped feed line and four parallel strips and have a very simple layout. The size of the structure is $50 \times 35 \times 1.6 \text{ mm}^3$. The quad band antenna shows frequency coverage of 2.78–2.91, 3.59–3.94, 5.15–5.33, and 5.39–6.06 GHz. For WLAN and WiMAX applications, a FR4-based compact tri-band antenna incorporating a split-ring resonator array is proposed in [9]. Its operating frequency ranges are 2.22–2.52, 3.42–3.68, and 5.3–5.65 with return losses of about -27, -38, and -15 dB, respectively, and gains of 2.25, 3.72, and 2.71 dB, respectively. A Rogers RT5880-based $40 \times 40 \times 0.79 \text{ mm}^3$ 16-element combination circular and rectangular patch array antenna has been introduced in [10]. Another diagonally connected 7-square element antenna having a bandwidth of 1.40 GHz and a maximum gain of 3.90 dB is described for WiMAX and 5G applications [12] respectively.

A tri-cell combo of rectangle slotted patch 1×4 array antenna with a return loss of -33.6 dB is designed in [16]. The VSWR corresponds to 1.02. The antenna is designed with $143 \times 71 \times 0.8 \text{ mm}^3$ FR4 substrate. Another ultra-wideband $36.25 \times 34.75 \times 0.79 \text{ mm}^3$ array antenna is presented in [14] for WLAN applications. The antenna uses some parasitic elements and is somewhat complex in structure. A dual-band (2.33–2.51 GHz and 5.09–5.92 GHz) antenna with directive radiation patterns is described in [4], where two symmetrical rectangular-shaped strips and F-shaped strips have been incorporated into the ground layer to get a suitable impedance match. L.C. Paul et al., proposed a pi-slotted antenna with a partial ground plane for 5G, Wi-Fi, and WiMAX applications [13]. The antenna shows a simulated return loss of -36.81 dB and a gain of 2.647 dB at the resonant frequency of the antenna. In [6], a 2.4 GHz slotted SIW array antenna for WLAN has been illustrated. The array antenna, comprised of a 12-slot SIW array, has diverse structures for the longitudinal slot array. Its dimension is $395.4 \times 120 \times 1.6 \text{ mm}^3$ with an operating range of 2.373–2.425 GHz and a return loss of about -31.813 dB.

The overall gain is 4.8 dB, and the value of the VSWR is less than 2. A simulation and analysis of $50 \times 60 \times 1.58 \text{ mm}^3$ antenna with a peak gain of 4.108 dB for WLAN applications is presented in [2]. The antenna has a return loss of -20.62 dB and an efficiency of 56.93%. A 4-element Roger RT5880 substrate array antenna resonating at 5.382 GHz and 6.864 GHz has been designed for WLAN applications. The array antenna also uses a T-junction in the feeding system to improve the radiation performance [11].

Therefore, after analyzing the literature review, there are still some research gaps that need further research and improvement. The demand for antenna shrinkage without sacrificing bandwidth is ever more urgent as the newest handy devices are getting smaller day by day. We are intending to design a miniaturized antenna capable of covering IEEE 802.11 a/h/j/n/ac/ax WLAN applications with a good reflection coefficient, gain, VSWR, and efficiency characteristics.

3 Design Structure

In this section, the design structure of the proposed low-profile wideband 1×2 array antenna has been discussed. The antenna has an overall dimension of $40 \times 22.4 \times 0.79 \text{ mm}^3$. Rogers RT5880 dielectric material with a thickness (H) of 0.79 mm, a relative permittivity of 2.2, and a tangent loss of 0.0009 acts as the antenna's substrate. The radiating patch is comprised of a 0.035-mm-thick copper plate. The uppermost portion of the proposed structure consists of a radiating patch in the form of a T-shaped strip and two circular patches, which are fed by a rectangular-shaped feed line. The T-junction deployed at the front-middle portion of the substrate. The T-junction strip has been used to improve the performance of the antenna. Where T-junction formation is utilized mainly for better impedance matching and also for enhancing the radiation characteristics of the antenna. A ground plane measuring $21.8 \times 22.4 \text{ mm}^2$ is fixed to the backside of the substrate. The ground plane is also 0.035 mm thick and made of copper. The design and simulation processes have been done using CST Microwave Studio. After modifying all of its physical characteristics and repeatedly evaluating its performance, the final design for the antenna's requirements has been chosen. The designed antenna's substrate is placed between the radiating patch and the ground plane.

A simple rectangular-shaped ($a \times b$) main feeder with a dimension of $22 \times 2.42 \text{ mm}^2$ is utilized. A horizontal rectangular intermediate strip with the dimension ($c \times e$) of $2 \times 14 \text{ mm}^2$ is introduced, which is illustrated in Figure 1(a). Another horizontal rectangular connecting strip is attached to the upper portion of the intermediate strip. Thereafter, two vertical rectangular sub-feeders are used to connect circular patches with the main feeder. Due to the connection with the rectangular strip, the return loss (dB) profile at both

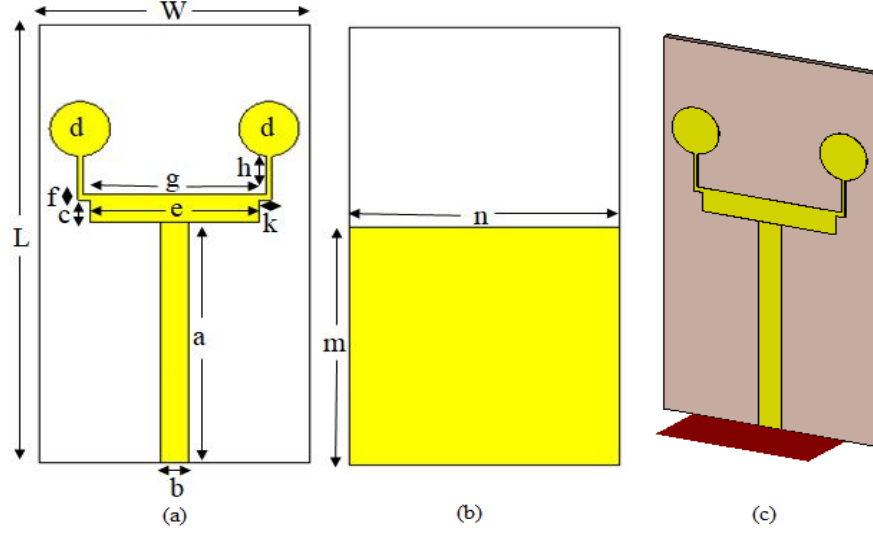


FIGURE 1: Proposed compact 1×2 array antenna for WLAN: (a) Front view, (b) Back view and (c) 3D view.

the center frequency as well as across the operational frequency range (GHz) of the antenna has been dramatically improved. This antenna is distinguished by the existence of two circular copper patches, fed by a unique T-junction, on the top side of the microstrip antenna. These two circular patches have the same radius (d), which is 2.5 mm. The connecting strip, which is attached to a horizontal rectangular intermediate strip, has a dimension ($f \times g$) of $0.5 \times 16 \text{ mm}^2$. The sub-feeders, which are introduced between the circular patches and the two ending points of the connecting strip, have an equal dimension ($h \times k$) of $3.51 \times 0.4 \text{ mm}^2$.

A ground plane with a dimension nearly half the size of the antenna is applied to the bottom side of the substrate. After adjusting all of its physical characteristics and repeatedly estimating its performance, the final length and width of the ground plane have been selected as presented in Figure 1(b). The entire dimension of the copper ground plane ($m \times n$) is $21.8 \times 22.4 \text{ mm}^2$. Figure 1(c) displays a 3D image of the proposed low-profile wideband 1×2 array antenna with its port after applying different standard techniques such as a T-junction, a rectangular feeder, two circular patches with sub-feeders, and a half-ground plane. The geometrical dimensions of the low-profile wideband 1×2 array antenna are listed in Table 1.1.

TABLE 1.1

List of parameters

Symbol	Value(mm)	Symbol	Value(mm)	Symbol	Value(mm)
L	40	W	22.4	H	0.79
t	0.035	a	22	b	2.42
m	21.8	n	22.4	c	2
e	14	d	2.5	f	0.5
g	16	h	3.51	k	0.4

4 Results and Analysis

The simulated performance of the designed low-profile wideband 1×2 array antenna is suitable for IEEE 802.11a/h/j/n/ac/ax WLAN applications, and it is presented in this section. The presented wideband 1×2 array antenna is characterized by simulation results and examined for its radiation properties based on the configurations specified in the above section. The CST-MWS suite is a high-performance 3D EM software that is used to construct, evaluate, and optimize the represented antenna. In Figure 2(a), the antenna's simulated reflection coefficient profile is displayed. The antenna is resonated at 5.672 GHz, which is its center frequency. The antenna covers the specifications for IEEE 802.11a/h/j/n/ac/ax WLAN frequency bands. It demonstrates the return loss of the proposed array antenna, which is -45.50 dB at 5.672 GHz. The operational frequency band of the antenna is between 4.72 GHz and 6.32 GHz. The array structure improves performance and boosts signal strength.

The Z-parameter is employed to calculate an antenna's quality factor, which can provide information on the achievable bandwidth. The impedance (Z) is an assertion of the resistance to alternating as well as direct electric current that a component of an electronic circuit or system offers. Resistance and reactance are two distinct scalar (one-dimensional) phenomena that make up the vector (two-dimensional) quantity known as impedance. Real and imaginary factors make up the Z parameter. The imaginary part of effective antennas should be near zero Ohm, and the real part needs to be close to fifty Ohm. The proposed antenna's Z parameter is shown in Figure 2(b). The Z parameter of the antenna has a real part of 49.67 and an imaginary part close to zero at the antenna's center frequency of 5.672 GHz. The voltage standing wave ratio, or VSWR for short, is a measurement of the effectiveness with which radio-frequency power is transferred from an energy source through a transmitting line into a load (for instance, from a power enhancer through a transmitting line to an antenna). Figure 2(c) displays the VSWR versus frequency curve of the designed array antenna. A very good VSWR of 1.01 is obtained at the proposed antenna's center frequency of 5.672 GHz. The VSWR profile complies with the criteria for an antenna design to have a VSWR that is close to unity. The VSWR is 1.93 at the lower cutoff point,

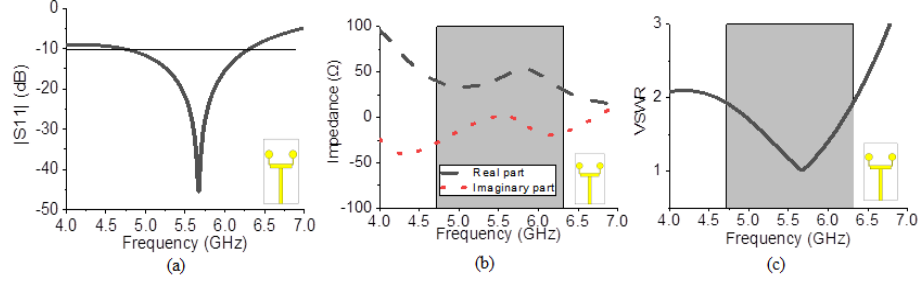


FIGURE 2: (a) Reflection coefficient profile, (b) Z-parameters and (c) VSWR within the operating range.

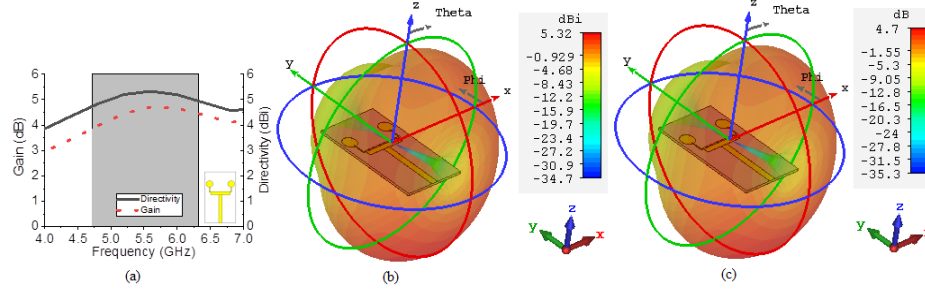


FIGURE 3: (a) Linear gain and directivity profile, (b) Directivity at 5.672 GHz and (c) Gain at 5.672 GHz.

where the frequency is 4.72 GHz. On the other hand, at the upper cutoff point, where the frequency is 6.32 GHz, the VSWR is 1.94. This indicates that the 5.672 GHz antenna covers the range of 4.72 GHz to 6.32 GHz for VSWR less than 2.

When compared to a reference antenna, an antenna's ability to concentrate radiated power or absorb incident radiation in a certain direction is referred to as antenna gain. The operational frequency range is depicted by the shaded area in Figure 3(a), which also shows the linear gain and directivity. With improved directivity (5.321 dBi) and gain (4.705 dB) at the center frequency of 5.672 GHz, the suggested antenna in this study makes an excellent candidate for application in IEEE 802.11a/h/j/n/ac/ax WLAN systems. The directivity range is 4.73–5.30 dBi and the gain range is 3.86–4.705 dB for the operating frequency range of 4.72–6.32 GHz, respectively. Figures 3(b) and 3(c) depict the 3D directivity and gain, respectively, at the center frequency of 5.672 GHz. These three-dimensional statistics at 5.672 GHz also show the antenna's directivity to be 5.321 dBi and gain to be 4.705 dB.

The ratio of the antenna's output power to its input power is known as antenna efficiency. Basically, an antenna's purpose is to efficiently radiate power

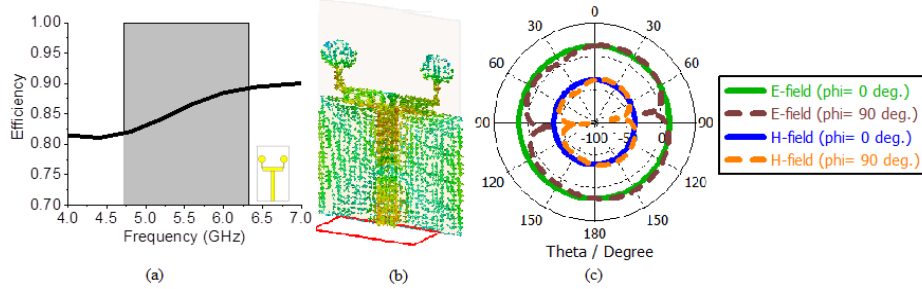


FIGURE 4: (a) Efficiency (b) Surface current distribution at 5.672 GHz and (c) Polar plot of both E and H-fields at 5.672 GHz.

that is applied at its input. The wideband 1×2 array antenna's simulated radiation efficiency is shown in Figure 4(a). An excellent simulated efficiency of 86.9% exists at the center frequency of 5.672 GHz. Efficiency is 81.61% at the lower cutoff point, where the frequency is 4.72 GHz. The efficiency is 89.90% at the upper cutoff point where the frequency is 6.32 GHz. The surface current distribution of the WLAN antenna at 5.672 GHz is presented in Figure 4(b). The current density is comparatively higher at the lower side of the T-junction area. The estimated maximum current density is 103.876 A/m at 5.672 GHz. The direction of the main lobe for both E and H fields is 180 degrees at $\phi = 0$ degree whereas it is 146 degrees at $\phi = 90$ degrees. At $\phi = 90$ degrees, the side lobe level is -1.3 dB. Angular widths are 123.1 and 60.4 degrees for $\phi = 0$ and 90 degrees, respectively, as presented in Figure 4(c). Table 1.2 provides a comparison of our proposed work with some recently published works. Our designed antenna has the minimum area and largest bandwidth with the highest efficiency in the IEEE 802.11a/h/j/n/ac/ax WLAN bands. From the table, it is clear that the proposed antenna possesses a good performance index, including a higher bandwidth, low volume, low return loss, and higher efficiency, which makes it suitable for the intended applications.

5 Conclusion

A wideband 1×2 array antenna for the advancement of IEEE 802.11a/h/j/n/ac/ax wireless local area networks has been presented in this research work. Utilizing a main feeder, a T-junction, two sub-feeders, and a partial ground plane, the radiation properties of the antenna are improved; its compact dimension is $40 \times 22.4 \times 0.79 \text{ mm}^3$. Furthermore, a partial ground plane (approximately half the area of the antenna) enhances the overall radiation characteristics, exhibiting a VSWR of 1.011, a return loss at the center frequency of -45.5

TABLE 1.2
Comparison table

Index	Reference No.								This work
	[2]	[3]	[4]	[6]	[7]	[8]	[9]	[16]	
Size (L×W×H) mm ³	50×60×1.58	50×35×1.6	50×69×1.6	395.4×120×1.6	210×150×1.6	50×40×1.6	32×37.2×1.6	143×71×0.8	40×22.4×0.79
Operating Frequency Range (GHz)	2.371-2.433	2.78-2.91, 3.59-3.94, 5.15-5.33, 5.59-6.06	2.33-2.51, 5.09-5.92	2.373-2.425	5.35-6.46	1.80-1.95, 5-5.16, 8.7-9.07, 12.40-15.12	2.22-2.52, 3.42-3.68, 5.3-5.65	~2.3-2.55	4.72-6.32
Centre Frequency (GHz)	2.40	2.83, 3.75, 5.24, 5.56	2.4, 5.5	2.4	5.8	1.85, 5.13, 9, 14.24	2.4, 3.5, 5.5	2.4	5.672
Reflection Coefficient (dB) at Centre Frequency	-20.62	~-23, ~-20.5, ~-27.5, ~-32	~-13, ~-22	-31.813	~-28	~-35, ~-20, ~-31, ~-15	~-27, ~-38, ~-15	33.6878	-45.5
Gain (dB)	Peak 4.108	-	6.55, 8.1	4.8	Peak 8.11	1.20, 2.66, 3.91, 6.6	2.25, 3.72, 2.71	-	Peak 4.705
-10 dB BW (GHz)	0.0632	0.13, 0.35, 0.18, 0.67	0.18, 0.83	0.052	1.11	0.15, 0.16, 0.37, 2.72	0.3, 0.26, 0.35	~0.25	1.60
VSWR	1.2054	-	-	~1.101	-	-	1.09, 1.02, 1.36	1.0233	1.01
Max. Efficiency (%)	56.93	-	70, 84.5	-	-	-	-	-	89.90
Application	WLAN	GPS, WLAN, WiMAX	WLAN	WLAN	WLAN	GSM 1800, WLAN, Radar, Direct Broadcast Satellite	WLAN, WiMAX	WLAN	IEEE 802.11a/h/j/n/ac/ax WLAN

dB, and a maximum radiation efficiency of 89.90%. In this design, a peak gain of 4.705 dB is obtained by employing a 1×2 array. The proposed antenna has an operating frequency range of 4.72-6.32 GHz with a center frequency of 5.672 GHz, and the structure has been constructed using a Rogers RT5880 substrate. This low-profile 1×2 array antenna would be an effective possibility for IEEE 802.11a/h/j/n/ac/ax wireless local area network.

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